

Vapor–liquid equilibria of propionic acid–water system in the presence of different types of inorganic salts: effect of temperature and salt concentration

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Abstract

Isothermal vapor–liquid equilibrium data at 40 and 50 °C were obtained by using a headspace gas chromatography (GC) (one of the static methods) for the propionic acid–water system in the presence and absence of chloride salts. Propionic acid was salted-out in the presence of any chloride salt, in the following order: aluminum chloride > calcium chloride > sodium chloride > ammonium chloride. The effects of AlCl₃, CaCl₂, and NaCl, at the three concentration levels 0.2, 0.5 and 1.0 m, revealed an increase in salting out of propionic acid with the increase of salt concentration. No remarkable effect of temperature on the VLE data and consequently on the enhancement factor of propionic acid (i.e. the ratio of relative volatilities with and without salt) was observed. The modified Furter correlation was successful in correlating the effect of dissolved salts on the VLE of the systems studied.

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1. Introduction

Most methods for separating azeotropic mixtures and low relative volatility mixtures rely on the addition of specially chosen chemicals to facilitate the separation. The problem of separating binary close-boiling and azeotropic solvent mixtures has lead to *enhanced* distillation techniques [1]. The basic methods to separate a binary azeotrope through distillation are extractive distillation, pressure-swing distillation, homogenous azeotropic distillation, heterogeneous azeotropic distillation, reactive distillation, and salt distillation [2]. This classification is based on the action of the additional component, the *entrainer*, which is introduced as an extra mass-separating agent. In salt distillation, the entrainer, salt, dissociates into ions in the liquid mixture and alters the relative volatilities sufficiently that the separation becomes possible.

Accurate vapor liquid-equilibria (VLE) experimental data are essential for the design of distillation columns. The presence of a salt may have a substantial influence on the system phase equilibria. In some cases, even small amounts of salt may bring about appreciable effects on the relative volatility of the solvent species in a liquid mixture. The addition of a salt, instead of a liquid, as a separating agent in distillation provides several advantages including lower energy consumption, and high purity of the overhead products because salts are non-volatile and hence do not evaporate or condense during distillation. The salt effect is believed to be a complex function of salt and solvent interaction and self-interaction among all other components [3–6]. In most cases, but not necessarily always, the molecules of the more polar solvent, in which the salt is more soluble, are preferentially attracted by the electrostatic field of the salt ions and thereby the molecules of the less polar solvent are liberated to the vapor phase (salted-out) [7].

The use of inorganic salts to alter the VLE of aqueous–alcohol systems has been extensively investigated. Reviews on the effects of inorganic salts on VLE

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are given by Furter and Cook [3] and Furter [4]. A compilation of VLE experimental data for many systems in the presence of salts is given by Ohe [8]. Various models are available in the literature to predict or to correlate the effect of salts on the VLE of mixed solvent–salt systems [9–14].

Propionic acid is one of the carboxylic acids that forms a minimum boiling point azeotrope with water at 0.053 acid mole fraction and 99.98 °C under atmospheric pressure. This acid is known to deviate from the ideal behavior even in the vapor phase where it forms dimmers [15]. Experimental data on the effect of inorganic salts on the VLE of the propionic acid–water system are limited [16]. The effects of a number of inorganic salts on the VLE of this system at 60 °C were presented in our previous works [17,18].

The aim of this work is, first, to present new VLE data sets with several salts including aluminum chloride, calcium chloride, sodium chloride, and ammonium chloride at 40 and 50 °C and, secondly, to show the effect of temperature and salt concentration on the VLE of the propionic acid–water system. The parameters of the modified Furter correlation recently proposed by Wu et al. [19] were regressed from VLE data obtained over a reasonable range of salt concentrations.

2. Experimental

The experiments for determining the VLE data of propionic acid–water–salt systems were conducted using 22 ml Pyrex glass vial filled with 3 ml of liquid samples of known concentration, a Genesis headspace autosampler and a Varian 3400 gas chromatography (GC). The GC was equipped with a capillary column (3 μ m, DB-624) and a flame ionization detector (FID). The temperature of the GC oven was 80 °C, ramped then to 18 °C. The injector and detector temperatures were 160 and 250 °C, respectively. A sample of the vapor phase formed in the glass vial was withdrawn by the auto-sampler and sent to the GC for analysis. The GC was calibrated by measuring peak areas for a set of standard solutions with known composition. Triplicate samples were prepared for each composition to ensure reproducibility, and the average values are presented here. The reproducibility was found to be within $\pm 1\%$. It was difficult to precisely measure the pressure inside vials because of the equipment limitations. Therefore, P – x , y diagrams are not presented. A detailed description of the experimental procedure is given elsewhere [17].

Distilled water and propionic acid of analytical grade (99.9% purity, supplier Sigma) were used. Salts used in this work had purities above 99%, and were dried when necessary, to remove moisture contents, before use.

To ensure the suitability of the experimental apparatus and the calibration methods, the experimental VLE

Table 1

VLE data of propionic acid (1)–water (2) at 40 °C with 0.5 m salt concentration

x_1	y_1					
		No salt	NaCl	NH ₄ Cl	CaCl ₂	AlCl ₃
0.05	0.035		0.049	0.036	0.049	0.067
0.10	0.062		0.083	0.072	0.093	0.119
0.20	0.091		0.132	0.115	0.139	0.161
0.30	0.120		0.160	0.153	0.170	0.183
0.40	0.149		0.172	0.176	0.182	0.203
0.50	0.159					
0.60	0.181					
0.70	0.207					

Table 2

VLE data of propionic acid (1)–water (2) at 50 °C with 1.0 m salt concentration

x_1	y_1					
		No salt	NaCl	NH ₄ Cl	CaCl ₂	AlCl ₃
0.05	0.031		0.070	0.063	0.081	0.145
0.10	0.058		0.115	0.110	0.135	0.167
0.20	0.095		0.170	0.161	0.176	0.205
0.30	0.125		0.230	0.185	0.230	0.254
0.40	0.133		0.252	0.230	0.250	0.310
0.50	0.151					
0.60	0.177					
0.70	0.217					

data for the salt-free propionic acid–water system at 40 and 50 °C obtained in this work were compared with those reported in the literature [8]. Since the results were in agreement with the literature experimental data, the experimental apparatus was concluded to be sound for studying salts effect on the VLE data.

3. Results and discussion

3.1. Effect of salt type

The effect of inorganic salts on altering the VLE of the propionic acid–water system at 40 and 50 °C was assessed experimentally using the salts: NaCl, NH₄Cl, CaCl₂, and AlCl₃ individually. The VLE data are presented in Tables 1 and 2 and plotted as y – x diagrams in Fig. 1A, Fig. 2A at 40 and 50 °C, respectively. The compositions shown in the tables and figures are on a salt-free basis. The results are also presented in Fig. 1B, Fig. 2B in terms of the enhancement factor, ϕ , defined as the ratio of the relative volatility of propionic acid in the presence of salt, α_s , to that under no-salt conditions, α_o ,

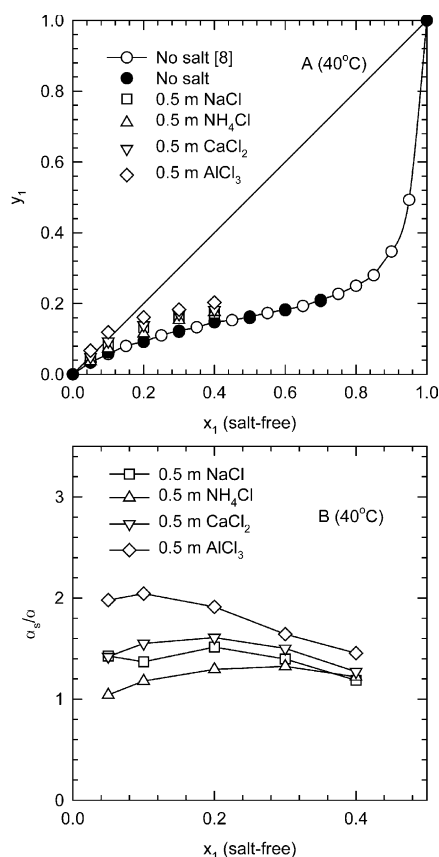


Fig. 1. The effect of several chloride salts on the VLE of the propionic acid (1)–water (2)-system at 40 °C.

$$\phi = \frac{\alpha_s}{\alpha_o} \quad (1)$$

where α is defined as:

$$\alpha = \frac{y_1/x_1}{y_2/x_2} \quad (2)$$

The subscripts 1 and 2 denote propionic acid and water, respectively. The enhancement factor is used as a measure of the salt effect strength exhibited by the concerned salt. When a salt is added to a liquid mixture it may exhibit salt-in and salt-out effects on the components. The salt-in effect reduces the volatility of one component while the salt-out effect increases that another causing an increase in the relative volatility. Therefore, if the enhancement factor of the concerned component is greater than 1.0 then it is concluded that that component has been salted-out while if the value is less than 1.0 then definitely that component has been salted-in. The enhancement factor is plotted as a function of propionic acid composition in the liquid phase, as shown in Fig. 1B, Fig. 2B at 40 and 50 °C, respectively.

As shown in Fig. 1A, Fig. 2A and as indicated by the enhancement factor values in Fig. 1B, Fig. 2B, propionic acid was salted-out in the presence of any one of the

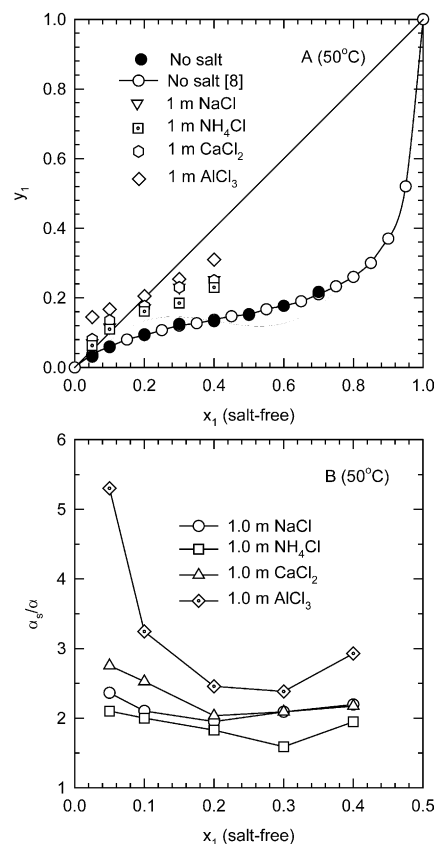


Fig. 2. The effect of several salts on the VLE of the propionic acid (1)–water (2)-system at 50 °C.

salts considered in this work. The addition of inorganic salts, which are preferentially soluble in water, increases the propionic acid content in the vapor phase, a behavior that is in conformity with the general phenomenon of salt effect. The degree of salting-out of propionic acid as indicated by the enhancement factor is salt type dependent. The salting-out of propionic acid was the largest with AlCl_3 and the smallest with NH_4Cl (Figs. 1 and 2). Based on the claim of electrostatic theories that salt effect should increase with a decreasing ion radius [20], this behavior should be expected since the ionic radius of Al^{3+} is 0.5 Å compared with 1.48 Å for NH_4^+ [21]. It can also be observed from Fig. 1B, Fig. 2B that the salt effect is dependent on the salt-free liquid composition. In the case of AlCl_3 (Fig. 2B), the enhancement factor decreased steadily from 5.3 to 2.5 as the mole fraction of propionic acid in the liquid phase increased from 0.05 to 0.2 and increased thereafter from 2.5 to about 3 as the liquid phase composition was raised from 0.2 to 0.4. The variation of the enhancement factor in response to the acid liquid phase composition in the case of NaCl , CaCl_2 , and NH_4Cl was less remarkable than that in the case of AlCl_3 (Fig. 2B).

Table 3

VLE data of propionic acid (1)–water (2) at 50 °C at different salt concentrations

x_1	NaCl			CaCl ₂			AlCl ₃		
	0.2 m	0.5 m	1.0 m	0.2 m	0.5 m	1.0 m	0.2 m	0.5 m	1.0 m
0.05	0.045	0.045	0.070	0.048	0.057	0.081	0.060	0.081	0.145
0.10	0.086	0.091	0.115	0.088	0.096	0.135	0.102	0.136	0.167
0.20	0.125	0.141	0.170	0.132	0.159	0.176	0.155	0.167	0.205
0.30	0.167	0.180	0.230	0.180	0.196	0.230	0.175	0.200	0.254
0.40	0.172	0.220	0.252	0.195	0.225	0.250	0.193	0.240	0.310

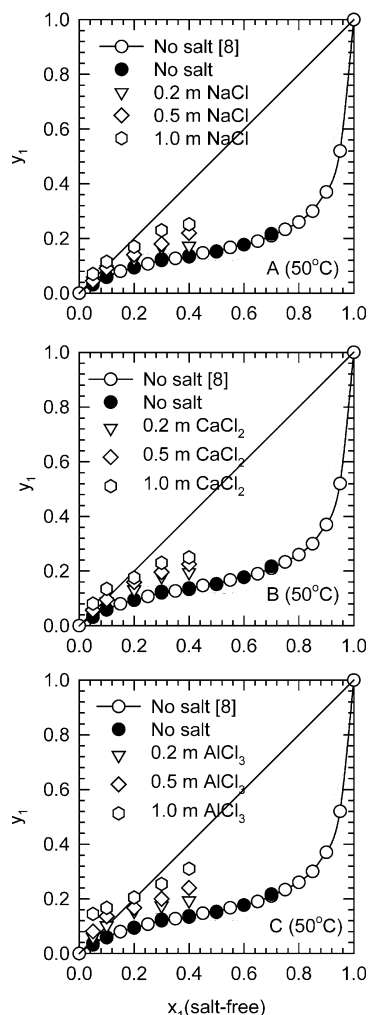


Fig. 3. The effect of salt concentration on the VLE of the propionic acid (1)–water (2)-system at 50 °C.

3.2. Effect of salt concentration

To point out the changes in the vapor-phase composition in the presence of different salt concentrations, the effects of AlCl₃, CaCl₂, and NaCl at three concentration levels, i.e. 0.2, 0.5 and 1.0 m, was studied. Experimental data on the VLE of propionic acid–water–salt systems at the three salt concentrations are presented in Table 3

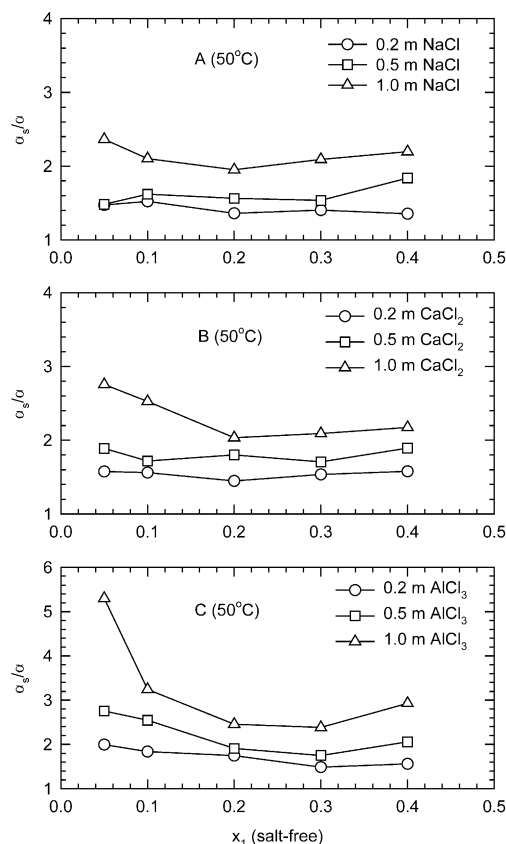


Fig. 4. Variation of the enhancement factor of propionic acid with salt concentration at 50 °C.

and plotted as y – x diagram in Fig. 3. These data were re-plotted in terms of the enhancement factor (Fig. 4) in order to analyze the salt effect.

As shown in Fig. 3, for all three salts, as the salt concentration increases in the liquid phase, the mole fraction of propionic acid increases monotonously in vapor phase, making the equilibrium line shift upwards. This trend is more obvious in Fig. 4, which shows the relationship between the enhancement factor and salt-free propionic acid mole fraction for NaCl (A), CaCl₂ (B) and AlCl₃ (C). It is seen that the enhancement factor notably increases with the increase in the salt concentration. For example, increasing the AlCl₃ concentration from 0.2 to 1.0 m at the acid mole fraction of 0.05

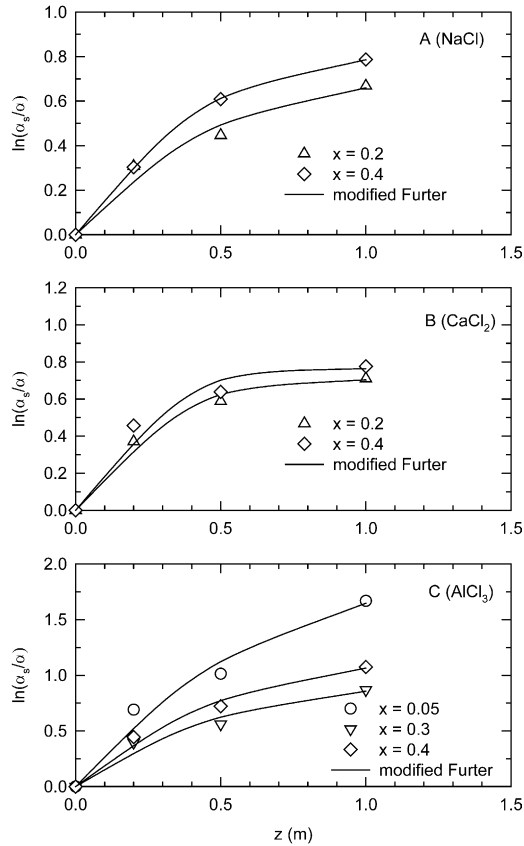


Fig. 5. Fitting of the experimental data to the modified Furter correlation. A, NaCl; B, CaCl₂; C, AlCl₃.

increased the enhancement factor from 2.0 to 5.3 (Fig. 4C). When the salts are dissolved in liquid mixture, they dissociate and exist in ionic form where they tend to attract preferentially the molecules of the more polar solvent and thus leaving the molecules of the less polar solvent to escape to the vapor phase. The solubility of the three considered salts in water, the more polar solvent, is greater than in propionic acid [16]. This could be due to the larger dielectric constant of water (80.4 at 20 °C) than that of propionic acid (3.1 at 15 °C). Therefore, the more salt is dissolved in the liquid mixture the more water molecules are attracted by the salt ions forming water–cation complexes and thus

weakening the acid–water interactions. As such, the escaping tendency of water is decreased by the better solvation power of this solvent while the escaping tendency of the acid is increased. This explains the increase in the enhancement factor of propionic acid with salt concentration. Similar findings were reported by Iliuta and Thyron [7] who found that salting-out effect on acetone increases with increasing NaSCN salt concentration in the acetone-methanol system. Yu et al. [22] observed a similar behavior for the effect of salt concentration on the vapor composition of formic acid when MgCl₂ was added to the formic acid–water system.

The effect of salt concentration on the enhancement factor when the composition of the solvent is held constant was first correlated by Furter [20] as:

$$\ln \phi = K_s Z \quad (3)$$

where K_s is called the ‘salt parameter’, and Z is the salt mole fraction. According to this equation, the salting-out effect exerted on propionic acid should increase with increasing salt concentration. Because of the limitation of this equation, especially at high salt concentrations, Wu et al. [19] modified the Furter equation as:

$$\ln \phi = k_1 Z + k_2 Z^2 \quad (4)$$

where k_1 and k_2 are two adjustable parameters. This modified Furter equation was used instead of the original Furter equation to correlate the enhancement factor of propionic acid because the later equation did not correlate well with the experimental data obtained in this work. The results of the three salts NaCl (Fig. 5A), CaCl₂ (Fig. 5B) and AlCl₃ (Fig. 5C) were well fitted to the modified Furter equation. Selected experimental results are presented in Fig. 5 to avoid confusion between the fitting curves. The values of the modified Furter correlation constants and the mean absolute deviations between the predicted and experimental values of the vapor mole fractions are given in Table 4.

Table 4
The values of the modified Furter correlation constants

x_1	NaCl			CaCl ₂			AlCl ₃		
	k_1	k_2	Dev.% ^a	k_1	k_2	Dev.% ^a	k_1	k_2	Dev.% ^a
0.05	1.165	−0.325	7.37	1.850	−0.851	5.23	2.842	−1.194	7.49
0.10	1.564	−0.838	6.20	1.635	−0.726	8.66	2.848	−1.684	4.68
0.20	1.312	−0.652	3.16	1.793	−1.089	3.27	2.163	−1.286	7.78
0.30	1.268	−0.543	4.71	1.731	−1.009	7.25	1.636	−0.779	4.37
0.40	1.664	−0.878	0.24	2.042	−1.278	5.84	2.022	−0.957	3.43

^a Deviation (%) = $(100/N) \sum (|y_1^{\text{cal}} - y_1^{\text{exp}}|/y_1^{\text{exp}})$.

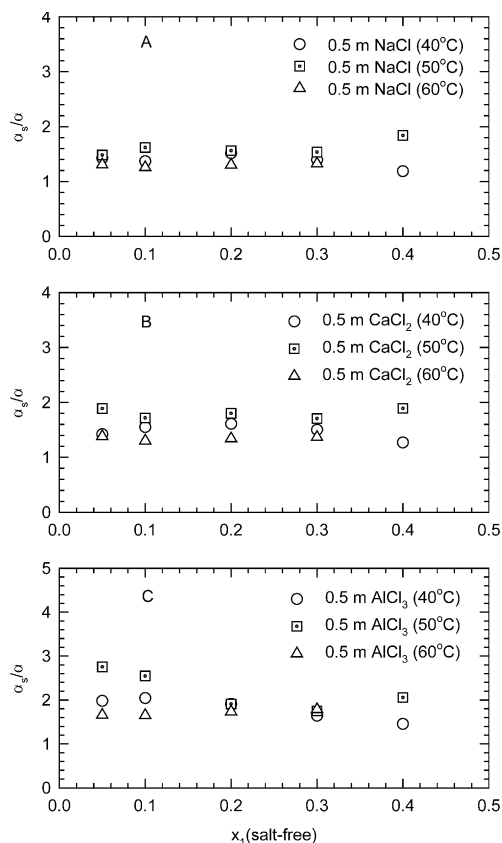


Fig. 6. Variation of the enhancement factor of propionic acid with salt temperature. A, NaCl; B, CaCl₂; C, AlCl₃.

3.3. Effect of temperature

Using the previous experimental data obtained at 40 and 50 °C, the enhancement factors of propionic acid at these temperatures along with the enhancement factors presented in earlier works at 60 °C [17,18] were plotted as a function of propionic acid concentration at a fixed chloride salt concentration of 0.5 m (Fig. 6). Equilibrium temperatures usually affect the mutual solubility between solvents, and salts solubility in the solvents mixture. Nonetheless, the variation of the enhancement factor within the temperature range studied was not considerable (Fig. 6). Therefore, the salt effect phenomenon was not seriously affected by temperature.

4. Conclusions

The VLE of the propionic acid–water system under no-salt condition and in the presence of four chloride salts (NaCl, NH₄Cl, CaCl₂, AlCl₃) dissolved to various concentrations were studied at 40 and 50 °C. The chloride salts used in this work have a salting-out effect on propionic acid in the following order: AlCl₃ > CaCl₂ > NaCl > NH₄Cl. The enhancement factor was

mainly dependent on the salt type and concentration, rather than on temperature. Increasing salt concentration led to the increase of salting-out of propionic acid. Experimental data were well correlated by the modified Furter equation.

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